

Harshith Mullapudi

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SUMMARY

Java Full Stack Engineer with 4+ years of experience building scalable microservices using Java, Spring Boot, Kafka, React, and AWS. Strong in designing event-driven architectures, optimizing APIs, and handling high-throughput distributed systems. Experience integrating AI/ML-driven services and NLP-based workflows into production systems to enhance automation and decision-making. Skilled in building reliable backend systems with a focus on performance, observability, and maintainability.

TECHNICAL SKILLS

- **Languages & Core Backend:** Java (8/11/17), Spring Boot, Spring MVC, Spring Security, Hibernate, REST APIs, GraphQL
- **Frontend & UI:** React.js, Redux, TypeScript, JavaScript (ES6+), HTML5, CSS3
- **Microservices & Distributed Systems:** Microservices Architecture, Service Decomposition, RESTful Services, API Design, Service-to-Service Communication, Event-Driven Architecture, Apache Kafka
- **Data & Persistence:** PostgreSQL, MySQL, MongoDB, Redis, Data Modeling, Query Optimization, Indexing Strategies
- **Cloud & Containerization:** AWS (EC2, EKS, S3, RDS), Docker, Kubernetes, Container Orchestration, Infrastructure Basics
- **CI/CD & DevOps:** Jenkins, GitHub Actions, Continuous Integration, Continuous Deployment, Build Automation
- **Testing & Quality:** JUnit, Mockito, Unit Testing, Integration Testing, Test Coverage Practices
- **Monitoring & Reliability:** Prometheus, Grafana, ELK Stack, AWS CloudWatch, Logging, Metrics, Alerting
- **Architecture & Design:** API Versioning, Fault Tolerance, Retry Mechanisms, Circuit Breaker Concepts, Scalable System Design
- **AI / Machine Learning:** NLP, Prompt Engineering, OpenAI APIs (basic usage), Vector Embeddings, RAG concepts, Scikit-learn (basic)

WORK EXPERIENCE

Software Engineer II

AT&T; | Dallas, TX | Oct 2024 – Present

Project: Intelligent Network Operations & Customer Experience Platform

- Developed Spring Boot microservices supporting telecom provisioning workflows, handling ~5K daily service activations across distributed systems.
- Designed REST and GraphQL APIs for communication across customer, billing, and partner services, improving system interoperability.
- Built Kafka-based event pipelines for provisioning and billing workflows, reducing cross-system data latency by 25%.
- Implemented Redis caching for high-frequency service metadata, improving API response times by 35% under peak load.
- Developed React dashboards aggregating microservices data, reducing manual tracking effort by 20% for operations teams.
- Secured APIs using Spring Security (OAuth2, JWT), ensuring role-based access control across internal and external systems.
- Optimized PostgreSQL queries and schema design, improving query performance during high concurrency scenarios.
- Containerized services using Docker and deployed on AWS EKS, enabling stable and repeatable production releases.
- Automated CI/CD pipelines using Jenkins and GitHub Actions, reducing deployment time by 40%.
- Integrated NLP-based log analysis to identify recurring failure patterns from ELK data, improving incident triage efficiency.
- Designed AI-assisted diagnostics using historical logs and event streams, improving root cause identification speed during production incidents.

Software Engineer

Kroger | Hyderabad, India | Jan 2022 – Dec 2023

Project: AI-Driven Retail Supply Chain & Customer Insights Platform

- Developed Spring Boot microservices for product catalog, pricing, and inventory systems supporting high-volume retail transactions.
- Designed REST APIs enabling communication between e-commerce, warehouse, and analytics systems in a distributed architecture.
- Built Kafka-driven pipelines for inventory synchronization, improving real-time data consistency by 20%.
- Delivered React dashboards visualizing inventory and demand data, enabling merchandising teams to make faster replenishment decisions.

- Integrated machine learning-based demand forecasting services, reducing stockout rates by 15%.
- Implemented Redis caching for high-frequency product queries, improving response time by 30% during peak traffic.
- Tuned PostgreSQL queries and indexing strategies, reducing latency during high-volume sales events.
- Deployed containerized applications on AWS EC2 and EKS, ensuring scalability and availability.
- Exposed ML model inference endpoints via REST APIs to support personalized recommendations in customer-facing applications.
- Built data pipelines to process user activity and feed recommendation systems using event-driven architecture.
- Applied fault-tolerant patterns (retries, circuit breakers) to improve resilience across distributed services.

PROJECTS

AI-Powered Customer Support Assistant

- Built a backend service integrating NLP-based APIs to classify customer queries and suggest contextual responses.
- Implemented embedding-based search to retrieve relevant knowledge base responses for customer queries.
- Designed asynchronous processing pipelines using Kafka to handle high-volume support requests.
- Exposed REST APIs for ticket ingestion, routing, and response generation.

Smart Recommendation Engine

- Developed microservices to process user activity streams and generate recommendations based on behavior patterns.
- Implemented event-driven processing using Kafka consumers for real-time recommendation updates.
- Applied basic machine learning techniques for user segmentation and recommendation improvement.
- Built low-latency APIs to serve recommendations to frontend applications.

ACHIEVEMENTS

- Improved API performance by 30% by introducing caching strategies and optimizing database queries across high-traffic endpoints.
- Reduced deployment time by 40% by implementing automated CI/CD pipelines for microservices.
- Increased system reliability by implementing event-driven communication, reducing data synchronization delays by 20%.
- Recognized for improving incident response workflows by enhancing observability and monitoring across distributed services.

EDUCATION

- **Master of Science in Artificial Intelligence** – University of North Texas

CERTIFICATIONS

- Oracle Certified Professional – Java SE 17 • AWS Certified Developer – Associate • Microsoft Azure Fundamentals (AZ-900)